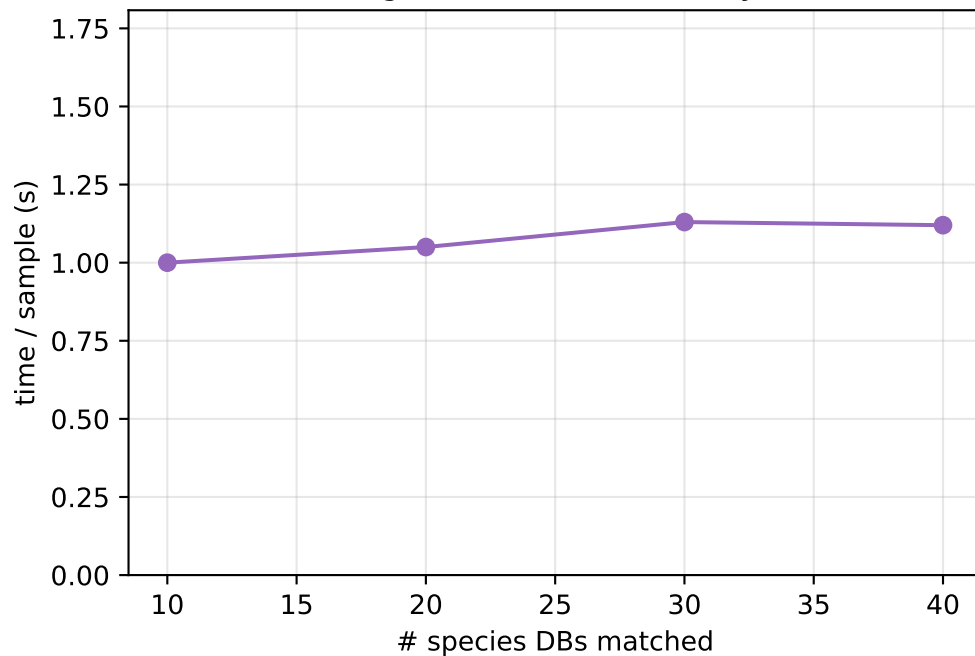
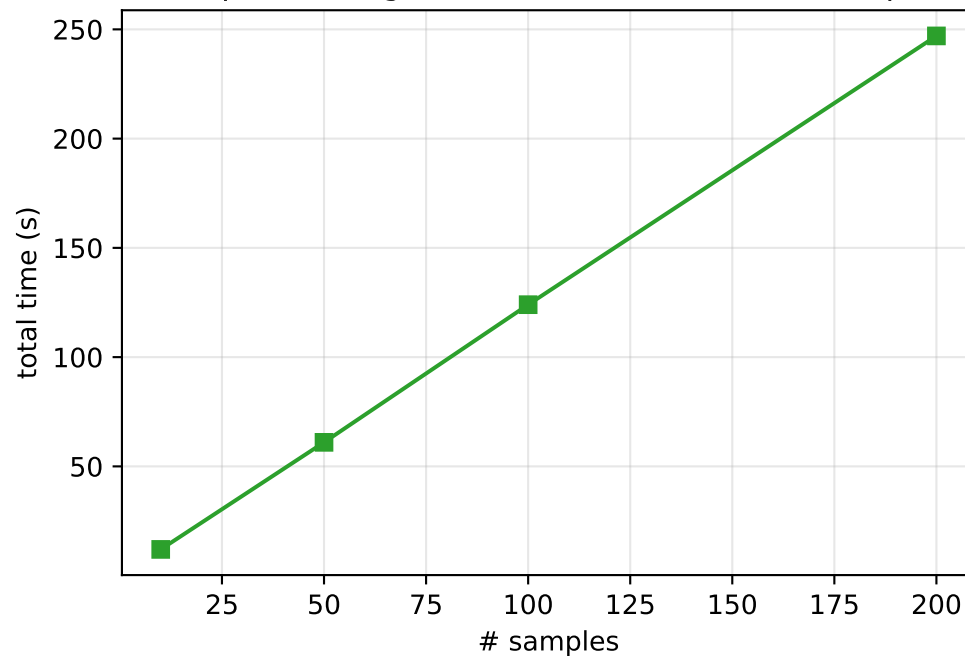


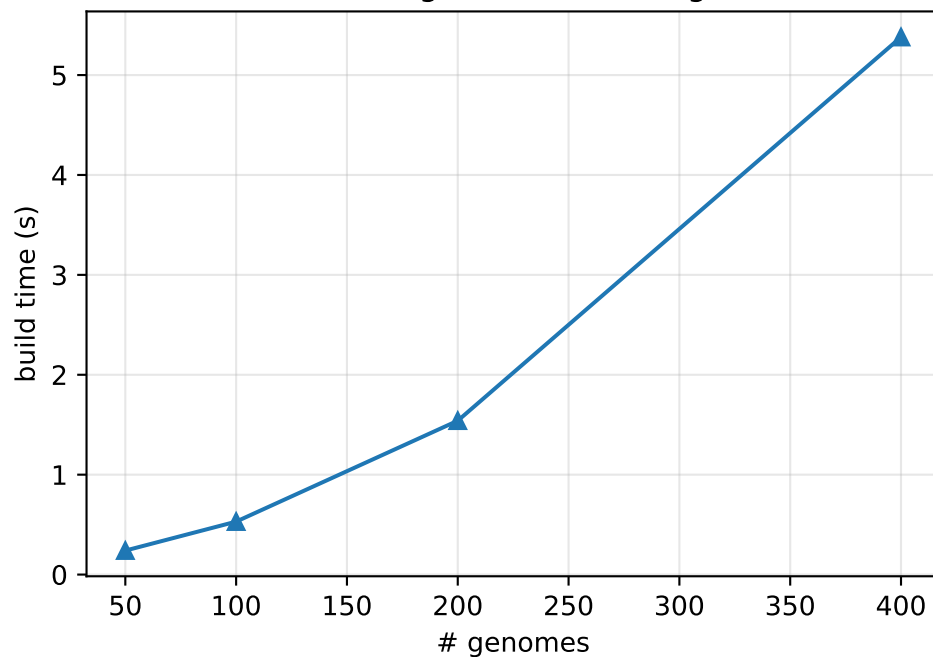
Species-count gradient — flat
(digest once, match many)



Sample-count gradient — linear (~1.24 s/sample)



DB-build scaling (MinHash >96 genomes)



Multi-species accuracy vs species gate
(recall stays 1.0)

